Supervised Learning (Part 3)

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Gradient

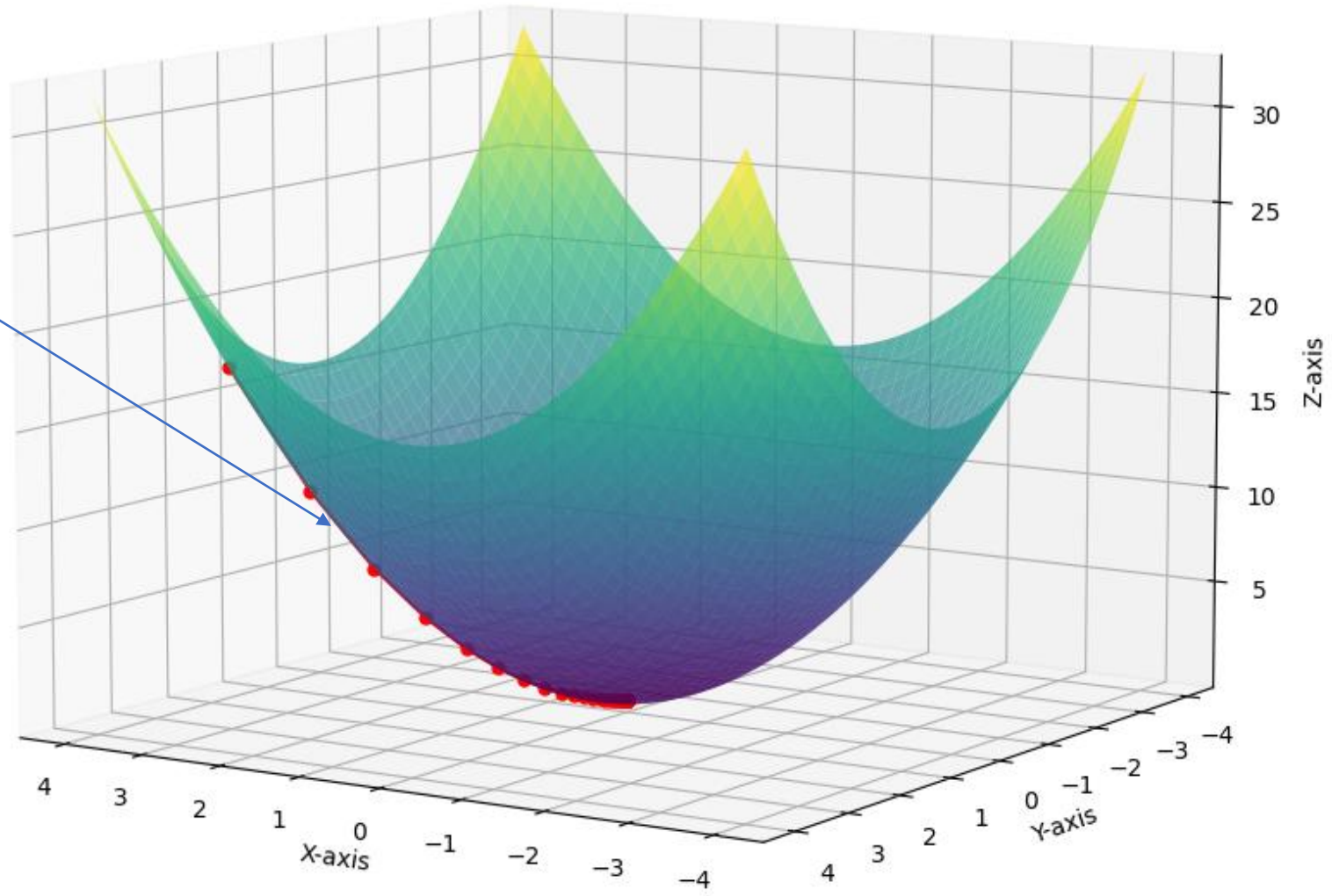
- The gradient of a scalar function is a vector that points in the direction of the greatest rate of increase of the function.
- The **gradient** of a function tells us two things:
  - **Direction of Steepest Ascent:** the direction in which the function increases the fastest.
  - **Rate of Increase:** the magnitude of the gradient tells how fast the function increases in that direction.

Gradient Descent

- **Intuition**: gradient tells us direction of greatest increase, negative gradient gives us direction of greatest decrease
  - If we take a small enough step in the direction of the negative gradient, the function will decrease in value
- **Goal**: to minimize MSE
  - Take the negative gradient to update parameter that reduces the error

Gradient of the function $x^2 + y^2$

Path taken by gradient descent, starting from the point [3.0,3.0]



Gradient Descent Algorithm:

1. Pick an initial point x_0
2. Gradient Descent Update Rule

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$

   - Where x_t is the current position at iteration t
   - α_t is the step size (learning rate)
   - $\nabla f(x_t)$ is the gradient (first derivative) of the function $f(x)$
3. Iterate until convergence

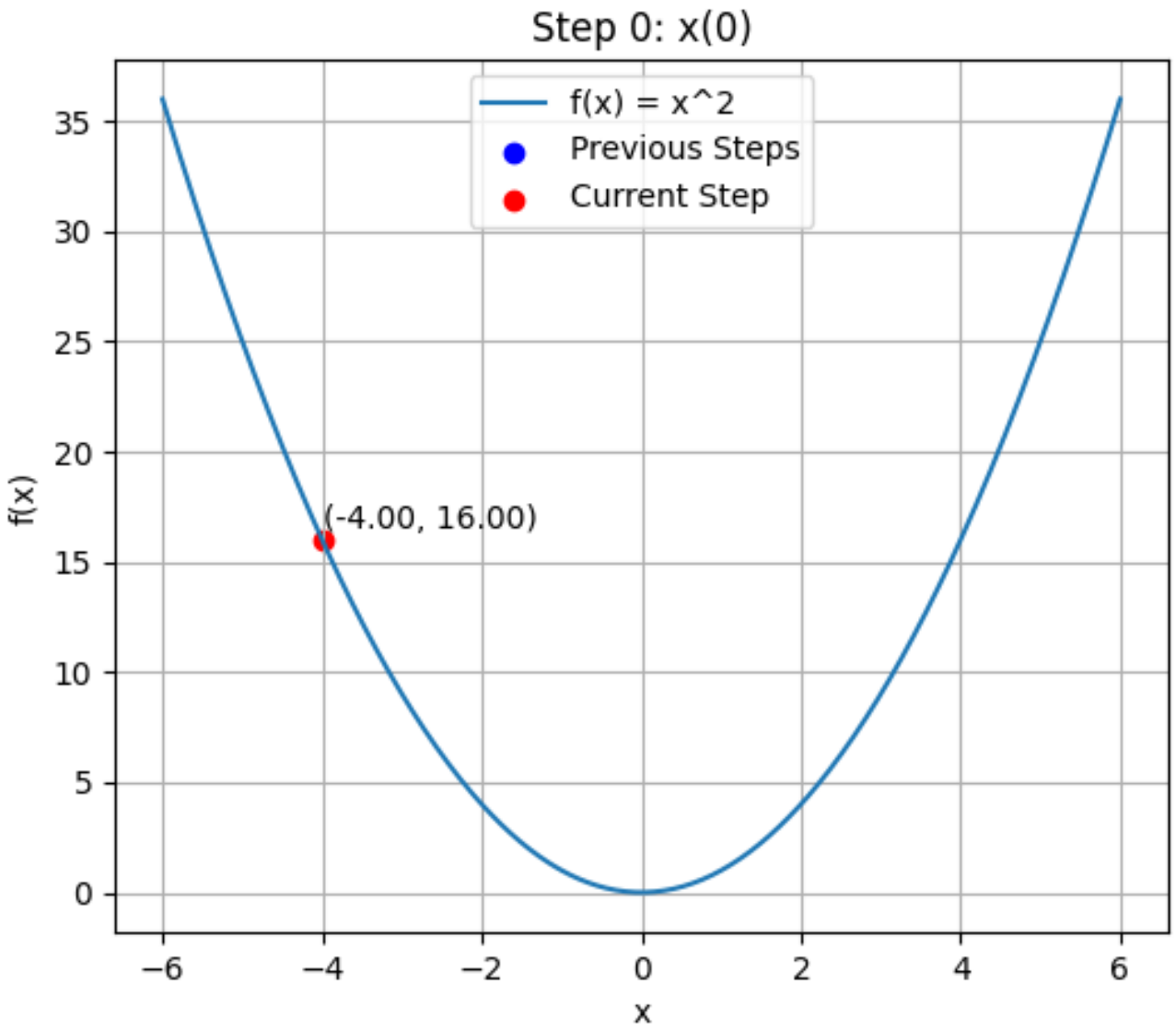
- $-\nabla f(x_t)$ moves in the negative direction of the gradient to minimize $f(x)$

Gradient Descent

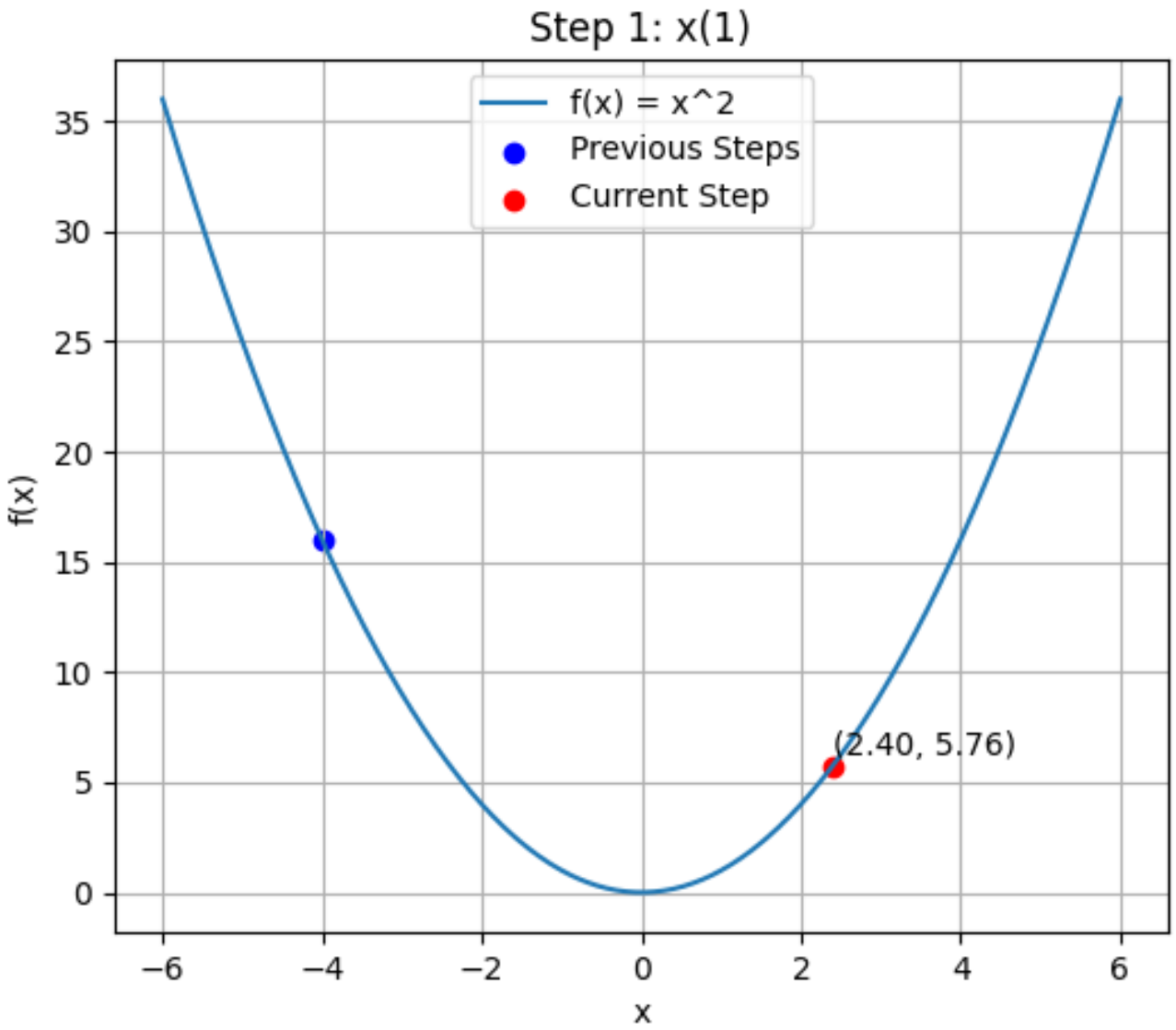
- When to stop?
  - Iterate until $\|\nabla f(x_t)\| \leq \epsilon$ for some $\epsilon > 0$
  - A typical choice of threshold, $\epsilon = 10^{-6}$
- How to choose step size α?
  - Try small values first (e.g., 0.01, 0.1, 0.5).
  - If gradient descent diverges (i.e., $f(x)$ keeps increasing), decrease α.
  - If convergence is too slow, try increasing α.
  - Instead of using fixed value, try using Line Search to find the best step size.

Gradient Descent

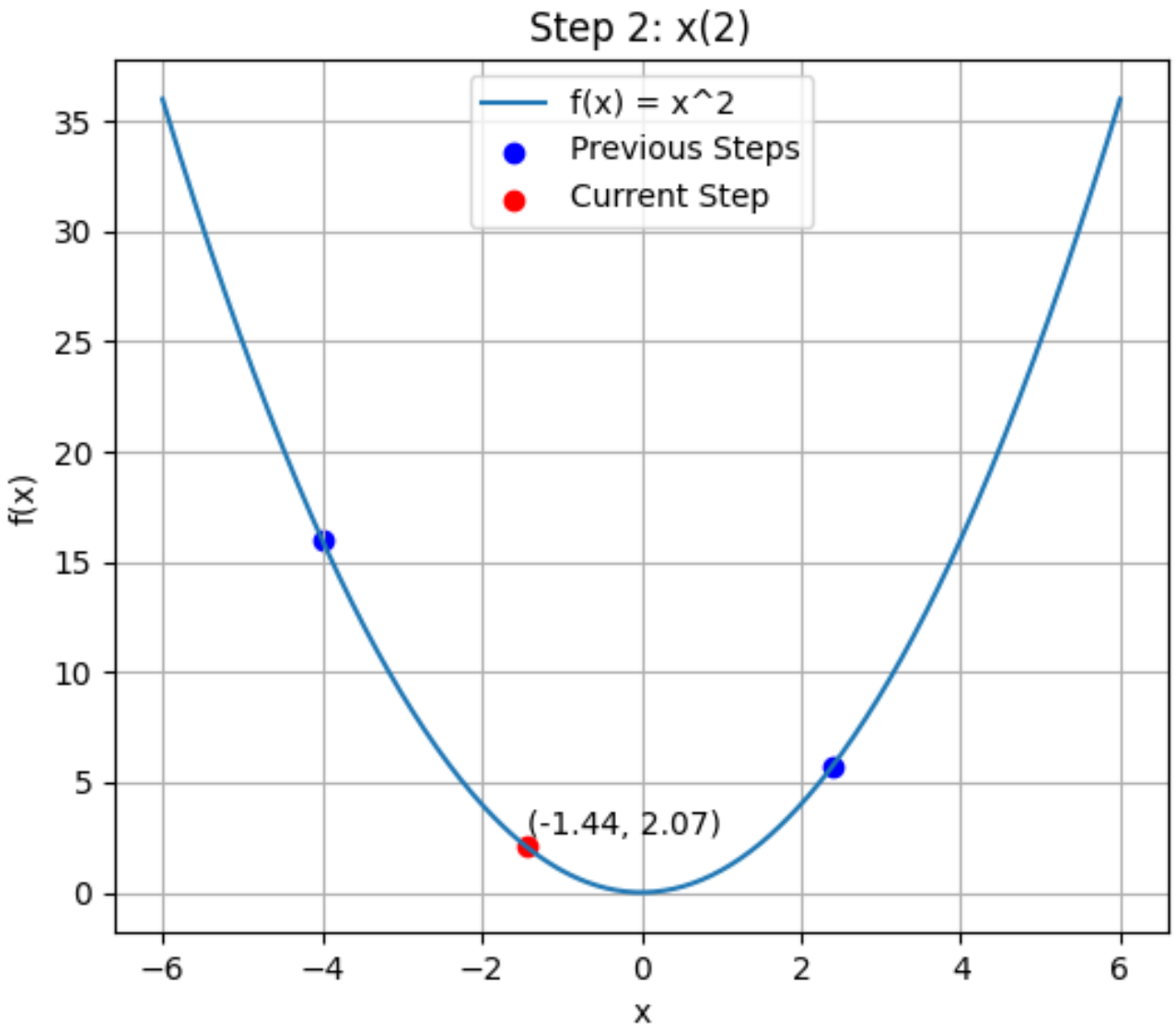
$f(x) = x^2$
Initial $x_0 = -4$
Step size, $\alpha = 0.8$



Gradient Descent



Gradient Descent



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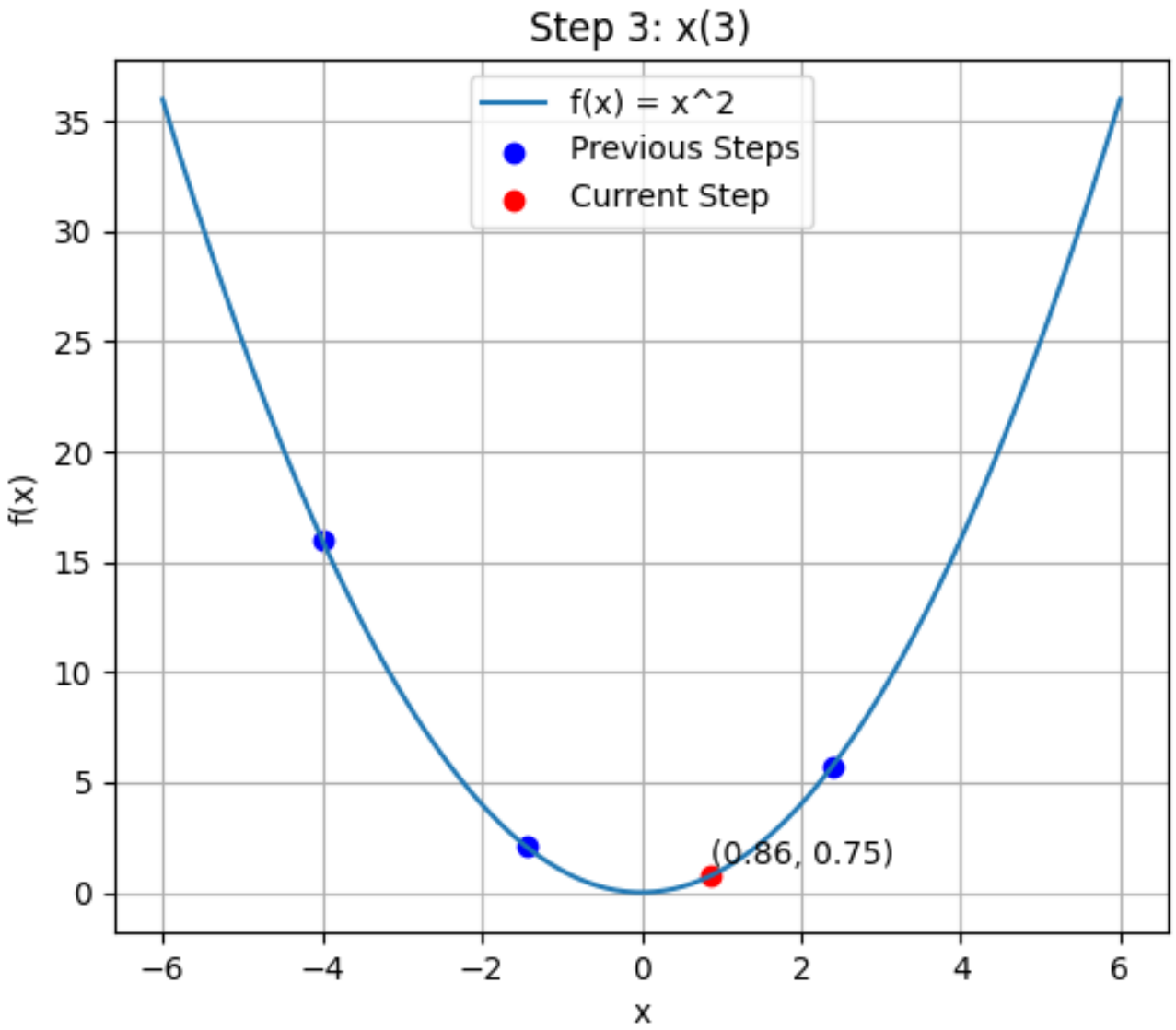

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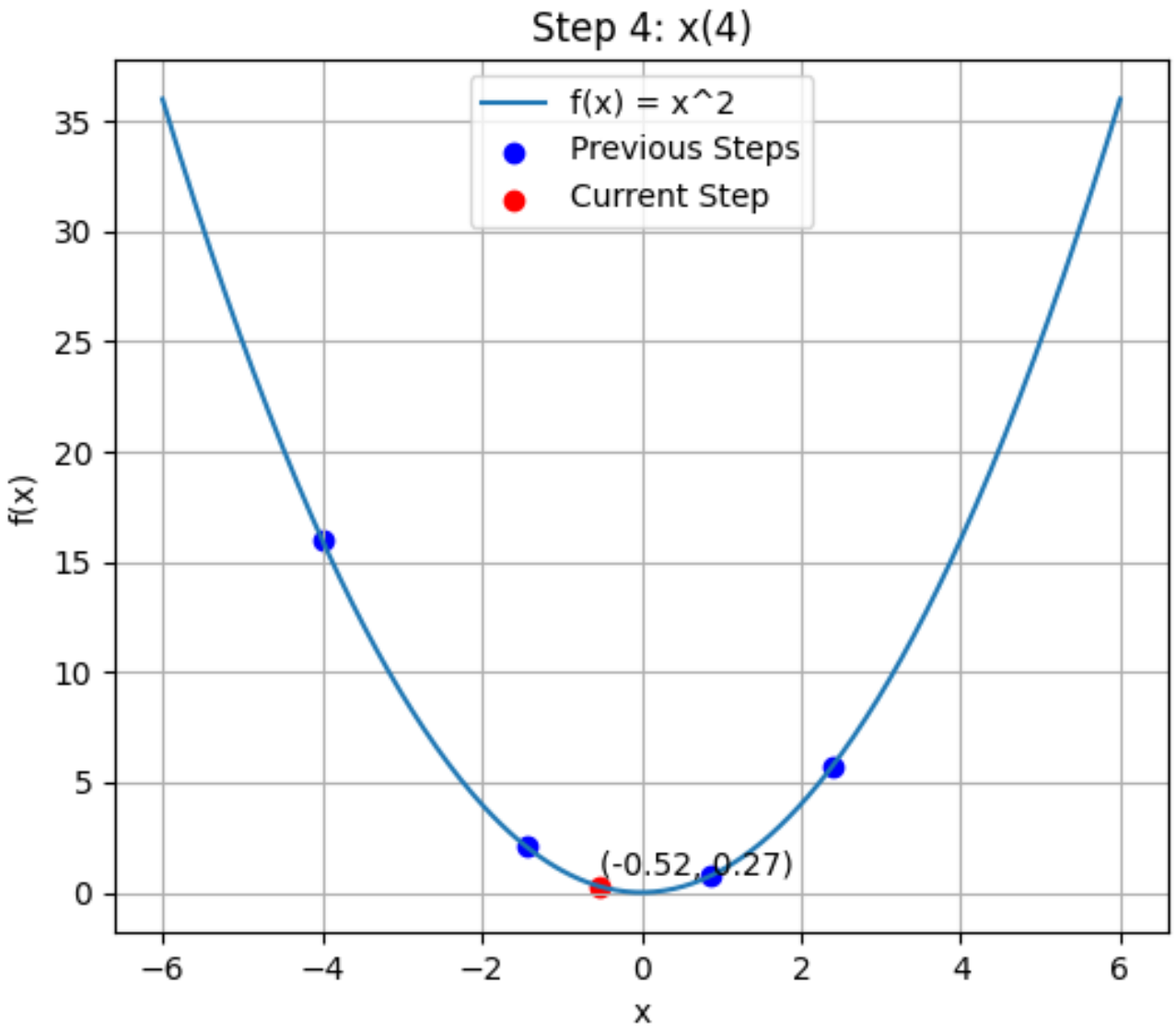
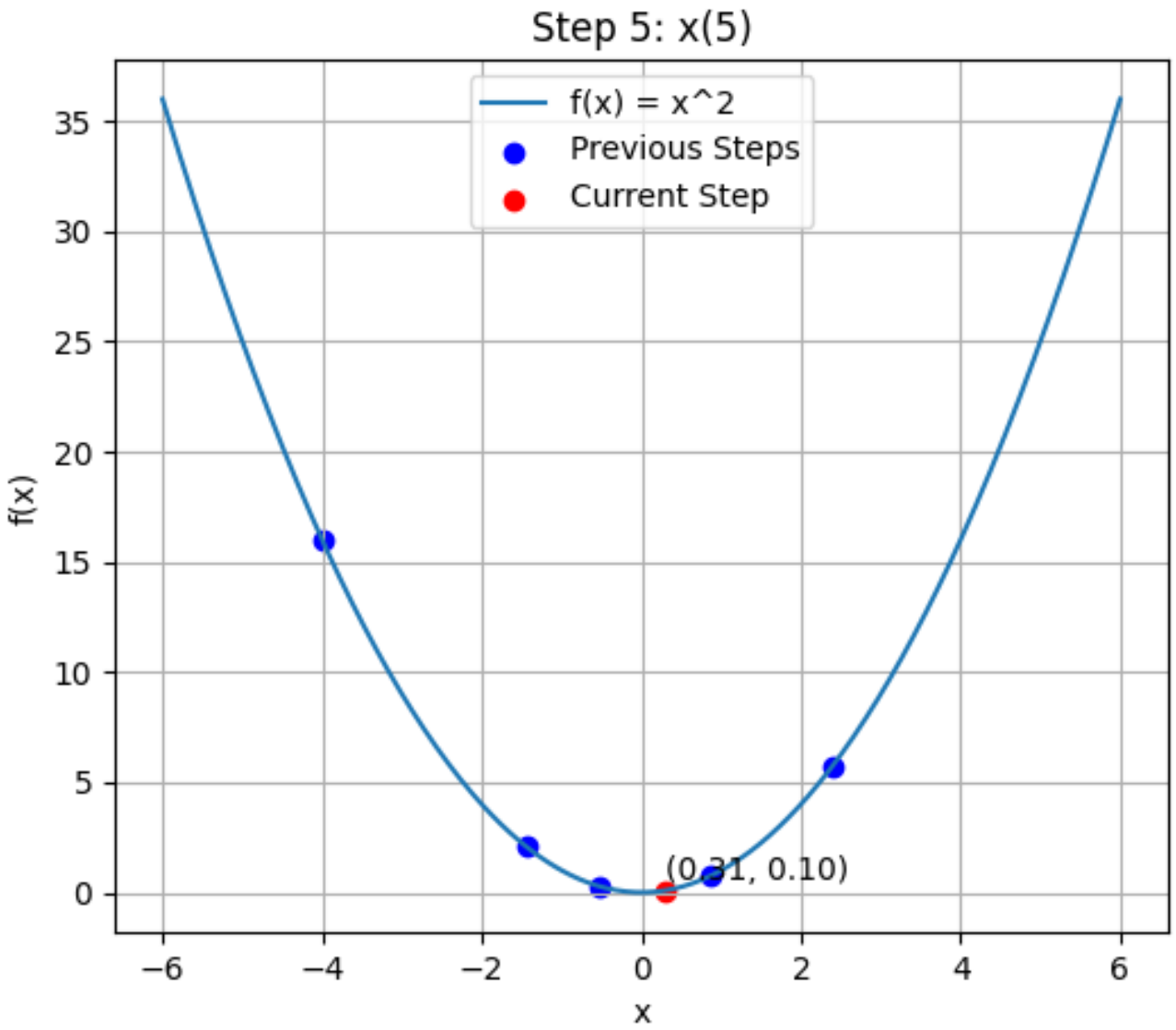


Gradient Descent



Gradient Descent for Linear Regression

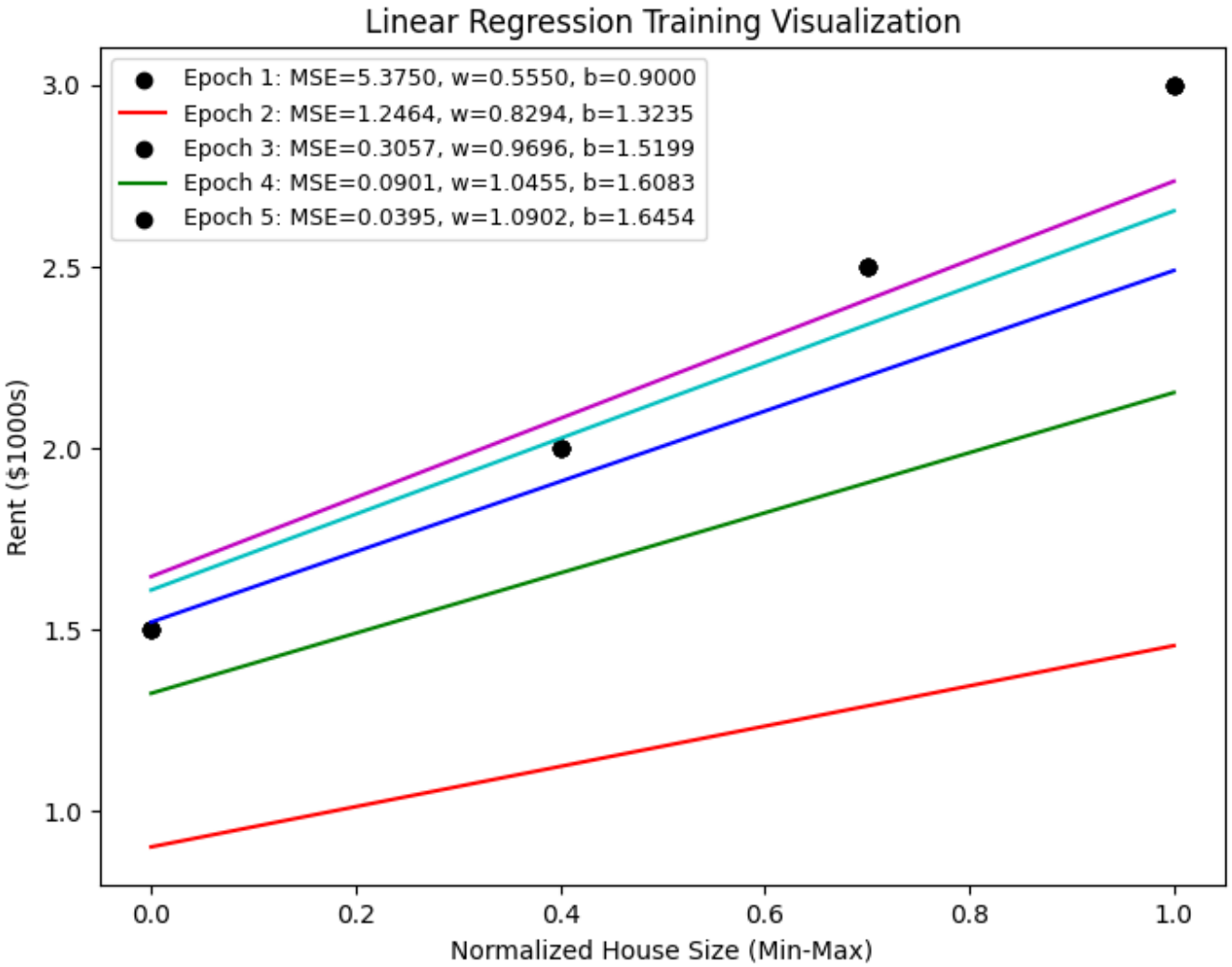
- Cost function: $J(W, b) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2$
- Update weight using Gradient Descent.
  - $W = W - \alpha\frac{\partial J}{\partial w}$
  - $\frac{\partial J}{\partial w} = -\frac{2}{n}\sum_{i=1}^{n} x_i(y_i - \widehat{y_i})$
  - $b = b - \alpha\frac{\partial J}{\partial b}$
  - $\frac{\partial J}{\partial b} = -\frac{2}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})$

Updated (-)

Fitting a Regression Line

House Size (Normalized)	Rent ($)
0.0	1.5
0.4	2.0
0.7	2.5
1.0	3.0

Starting with w = 0, b = 0, α = 0.2



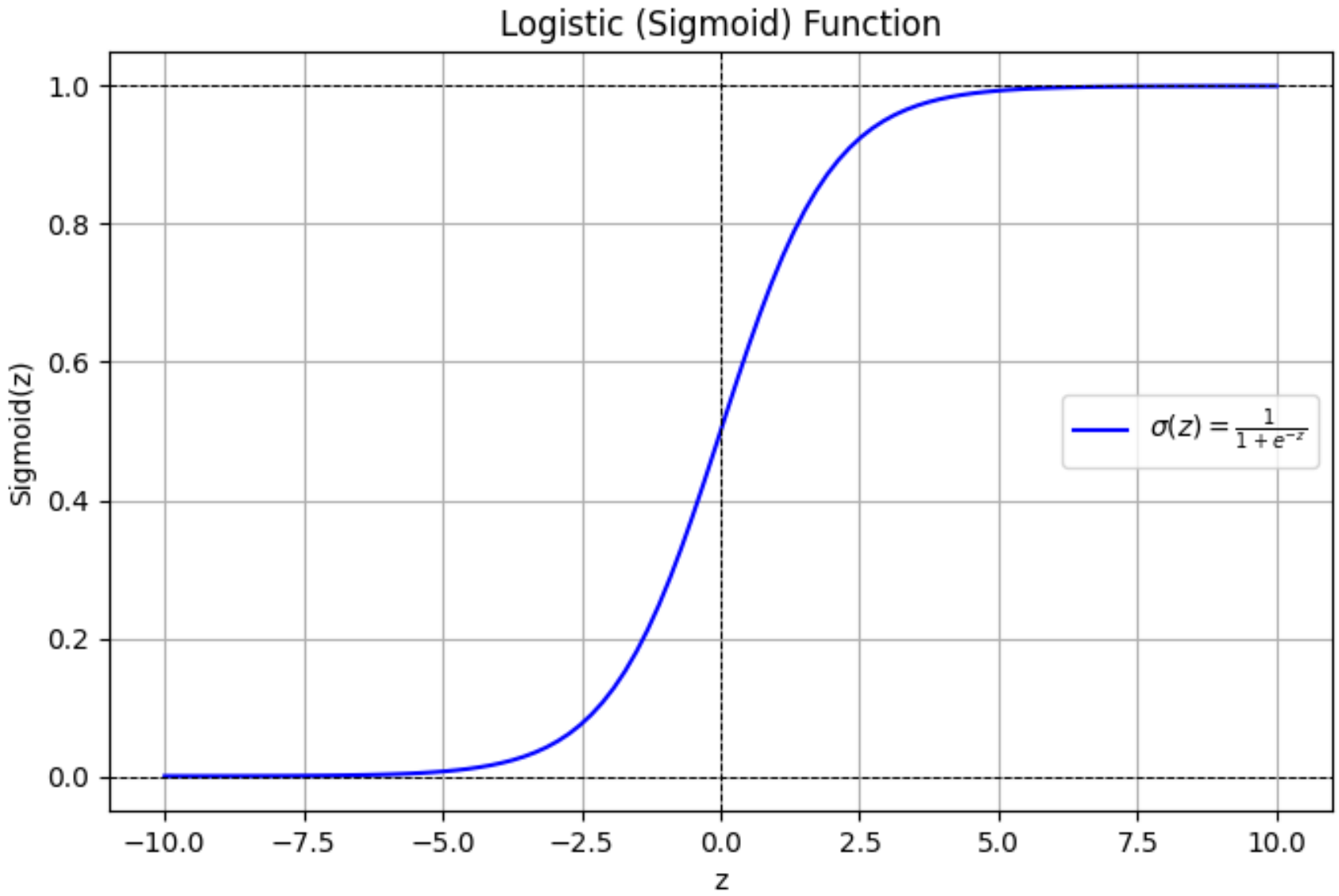
Logistic Regression

- Logistic regression is used for classification!!
- Predicts the probability of a binary outcome (0 or 1)
- Extends the idea of a linear regression to classification tasks

Logistic Function

- The logistic function (also called the sigmoid function) is used in logistic regression to model the probability of a binary outcome
- $\sigma(z) = \frac{1}{1+e^{-z}}$
- The logistic function turns a real number into a probability distribution.
  - If $\sigma(z)$>0.5, classify as 1
  - Else, classify as 0

Logistic Function



Linear Regression vs Logistic Regression

Key Differences:

- **Linear Regression**: Predicts continuous outcomes, assumes a straight-line relationship between the independent and dependent variables.
- **Polynomial Regression**: Extends linear regression to model non-linear relationships by using polynomial terms of the independent variable.
- **Logistic Regression**: Predicts probabilities for binary outcomes, using the logistic function (sigmoid) to squeeze the output between 0 and 1. It is used for classification rather than regression.

Questions?